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QUALITY MANAGEMENT SYSTEM

STANDARD OPERATING PROCEDURE

Contamination Control and Product Cleaning

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Current Version / Revision	00
Effective Date	[Date of Final Approval]
Review Cycle	Annual (Every 12 Months)
Applicable Regulatory Standards	ISO 13485:2016 Clause 6.4 & 7.5.2 ISO 14644 Series (Cleanrooms) FDA 21 CFR Part 820.70(c) & (e)

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1.0 Purpose

This Standard Operating Procedure (SOP) establishes the mandatory requirements for environmental contamination controls, gowning disciplines, chemical cleaning procedures, and particulate mitigation strategies within medical device manufacturing zones. The primary intent is to limit particulate, chemical, and microbial bioburden levels on product surfaces to safeguard subsequent sterilization validation states and prevent material pyrogenicity or chemical toxicity in clinical end-users.

2.0 Scope

This procedure applies to all cleanroom manufacturing environments (specifically ISO Class 7 and Class 8 infrastructure), in-process component chemical baths, ultrasonic wash stations, assembly tooling interfaces, and operator hygiene protocols. It governs all cleaning interventions applied to devices before final packaging and downstream terminal sterilization fields.

3.0 Responsibility and Authority

3.1 Cleanroom Operators / Production Technicians

Responsible for executing daily equipment cleaning tasks, adhering to standardized cleanroom gowning behaviors, and operating line chemical wash stations exactly as specified by batch records.

3.2 Environmental Monitoring (EM) Technician

Responsible for executing scheduled environmental monitoring audits, collecting active/passive air microflora metrics, taking surface contact samples, and validating cleanroom differential pressures.

3.3 Quality Assurance / Validation Engineer

Holds authority to verify washing bath parameters, audit gowning compliance, analyze particulate/microbial trend anomalies, and approve cleaning validation technical files.

4.0 Procedure

4.1 Controlled Environment Parameters Matrix

Manufacturing cleanrooms must maintain strict structural configurations monitored continuously or per lot run. Operational ranges are restricted to the following boundaries:

Cleanroom Zone	ISO Standard Level	Differential Pressure Target	Max Particle Limits (≥0.5 μm / \$m^3\$)
Core Assembly Room	ISO Class 7 (In Operation)	>15 Pascals vs. Ante-Room	352,000
Ante-Room / Gowning Box	ISO Class 8 (In Operation)	>10 Pascals vs. Uncontrolled	3,520,000
Material Air-Lock (MAL)	ISO Class 8 (In Operation)	Interlocked pressure buffers	3,520,000

4.2 Component Ultrasonic Cleaning Workflow

To eliminate processing oils, particulate matter, and chemical residues from mechanical interfaces, a multi-stage automated wash line must be operated sequentially:

Stage 1: Detergent Pre-Wash

Immerse components in a validated 2% v/v solution of analytical-grade non-ionic surfactant (e.g., Alconox or Liquinox) formulation. Maintain the bath temperature at 55°C ± 5°C with ultrasonic frequency sweep toggled at 40 kHz for a continuous duration of 10 minutes.

Stage 2: Primary Deionized Water Rinse

Transfer load into a flowing Deionized (DI) water cascading overflow basin. Ultrasonic agitation must remain active for 5 minutes to fully strip surfactant residues from internal porous channels or component geometric undercuts.

Stage 3: Critical Final Purified Water Flush

Execute a final rinse cycle using high-purity Purified Water (PW) conforming to current pharmacopeia thresholds. Water conductivity must monitor at <1.3 μS/cm at 25°C. Dwell exposure time is fixed at 5 minutes.

CRITICAL COMPLIANCE NOTICE: Chemical cleaning baths must be dumped and completely refreshed either every 24 hours of operational deployment or after treating a maximum defined surface area footprint. Operating in a saturated or exhausted cleaning bath will lead to contaminant redeposition, causing immediate lot rejection.

4.4 Operator Gowning Discipline and Personal Constraints

Personnel must pass through a mandatory step-by-step gowning sequence prior to crossing the cleanroom boundary line. Gowning artifacts include non-linting coveralls, hood ensembles, dedicated cleanroom boots, and powder-free nitrile gloves. No personnel shall bring personal cosmetics, jewelry, wristwatches, or unapproved paper logs into the controlled cleanroom enclosure.

4.5 Cleaning Validation Protocols (TOC and Bioburden Limits)

Cleaning protocols must be validated to confirm the removal of residues to non-toxic levels. Validation runs (requiring 3 consecutive processing batches) must meet objective passing limits before equipment release:

- **Total Organic Carbon (TOC):** Extraction rinse fluid samples must verify at <500 ppb or meet calculated maximum allowable carryover (MACO) boundaries.
- **Bacterial Endotoxins (Pyrogens):** Finished component fluid flushes must monitor at <20 EU/device for standard systemic contact options, and <2.15 EU/device for ophthalmic or neurological interface configurations.
- **Particulate Matter Cleanliness:** Microscopic checks must show particulate populations are well below standard inspectorate targets (e.g., complying with USP <788> fluid limits).

5.0 Documentation and Log Archiving

Washing bath parameter logs, cleaning solution batch tracking sheets, daily cleanroom pressure tracking sheets, and weekly microflora settle-plate logs are managed as critical Device History Record documents. These sheets are archived securely for a minimum of ten (10) years, or as dictated by global device technical files.

6.0 Associated Reference Materials

- **SOP-MF-010:** Sterilization Process Management and Validation
- **SOP-QA-012:** QMS Process and Equipment Validation Framework
- **Form-MF-011A:** Daily Cleanroom Parameter Log Sheet
- **Form-MF-011B:** Wash Station Solution Prep Tracking Log

7.0 Revision History Log

Revision Level	Effective Date	Author / Originator	Detailed Description of Modifications
00	[Date]	[Name], Manufacturing Ops	Initial baseline release of formalized Contamination Control and Product Cleaning Standard Operating Procedure.